

IGNOU BCS-011 / BCS-111**Computer Basics and PC Software**

Solved Question Paper: Term-End Examination, December 2020

*Full solution with concise explanations, tables, flowcharts and TikZ diagrams***Contents**

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Paper instructions

Question No. 1 is compulsory and carries 40 marks. Attempt any three questions from the rest. This solution booklet provides answers to all questions for practice.

1 Question 1

Question. (a) Perform the following conversions: (i) $(16.25)_{10}$ to binary; (ii) $(1011\ 1010)_2$ to decimal; (iii) $(ABC)_{16}$ to binary; (iv) $(110011\ 1001)_2$ to hexadecimal. *[6 marks]*

Answer.

(i) $(16.25)_{10}$ to binary:

$$16_{10} = 10000_2, \quad 0.25_{10} = \frac{1}{4} = 2^{-2} = 0.01_2.$$

Hence,

$$(16.25)_{10} = (10000.01)_2.$$

(ii) $(1011\ 1010)_2$ to decimal:

$$\begin{aligned} (10111010)_2 &= 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0. \\ &= 128 + 32 + 16 + 8 + 2 = 186. \end{aligned}$$

Therefore,

$$(1011\ 1010)_2 = (186)_{10}.$$

(iii) $(ABC)_{16}$ to binary: Convert each hexadecimal digit into four binary bits.

$$A = 1010, \quad B = 1011, \quad C = 1100.$$

Therefore,

$$(ABC)_{16} = (1010\ 1011\ 1100)_2.$$

(iv) $(110011\ 1001)_2$ to hexadecimal: Group binary digits from the right in groups of four.

$$1100111001_2 = 0011\ 0011\ 1001_2.$$

$$0011 = 3, \quad 0011 = 3, \quad 1001 = 9.$$

Therefore,

$$(110011\ 1001)_2 = (339)_{16}.$$

Question. (b) List the input/output devices for the following situations. Also describe the features of the device: (i) taking photographs, (ii) producing output of photograph, (iii) input of voice, (iv) output of voice. *[4 marks]*

Answer.

Situation	Device	Features
Taking photographs	Digital camera / webcam / mobile camera	Captures images using an image sensor; resolution is measured in megapixels; can store images digitally; may support zoom, autofocus and video recording.
Producing output of photograph	Photo printer / colour inkjet printer	Produces hard-copy photo output; supports colour printing, high resolution, glossy paper and accurate colour reproduction.
Input of voice	Microphone	Converts sound waves into electrical/digital signals; used for recording, voice chat, dictation and speech recognition.
Output of voice	Speaker / headphone	Converts digital audio signal into sound; used for music, voice calls, alerts and multimedia output.

Question. (c) What is the role of main memory in a computer? Why is it called RAM? How is RAM different from ROM? [4 marks]

Answer. Main memory stores the data, instructions and intermediate results that the CPU needs immediately while executing programs. It is much faster than secondary storage, so the CPU can quickly read and write information during processing.

It is called **RAM** because RAM means **Random Access Memory**. Any memory location can be accessed directly in almost the same time, without reading all previous locations sequentially.

Basis	RAM	ROM
Full form	Random Access Memory	Read Only Memory
Nature	Mostly volatile	Non-volatile
Use	Holds running programs and current data	Stores firmware or boot instructions
Read/write	CPU can read and write frequently	Mostly read-only in normal use
Effect of power off	Contents are lost	Contents remain stored

Question. (d) You have attached a new printer to your computer. Which software will you require so that the printer is installed correctly and starts to print? List features of the required computer software. [3 marks]

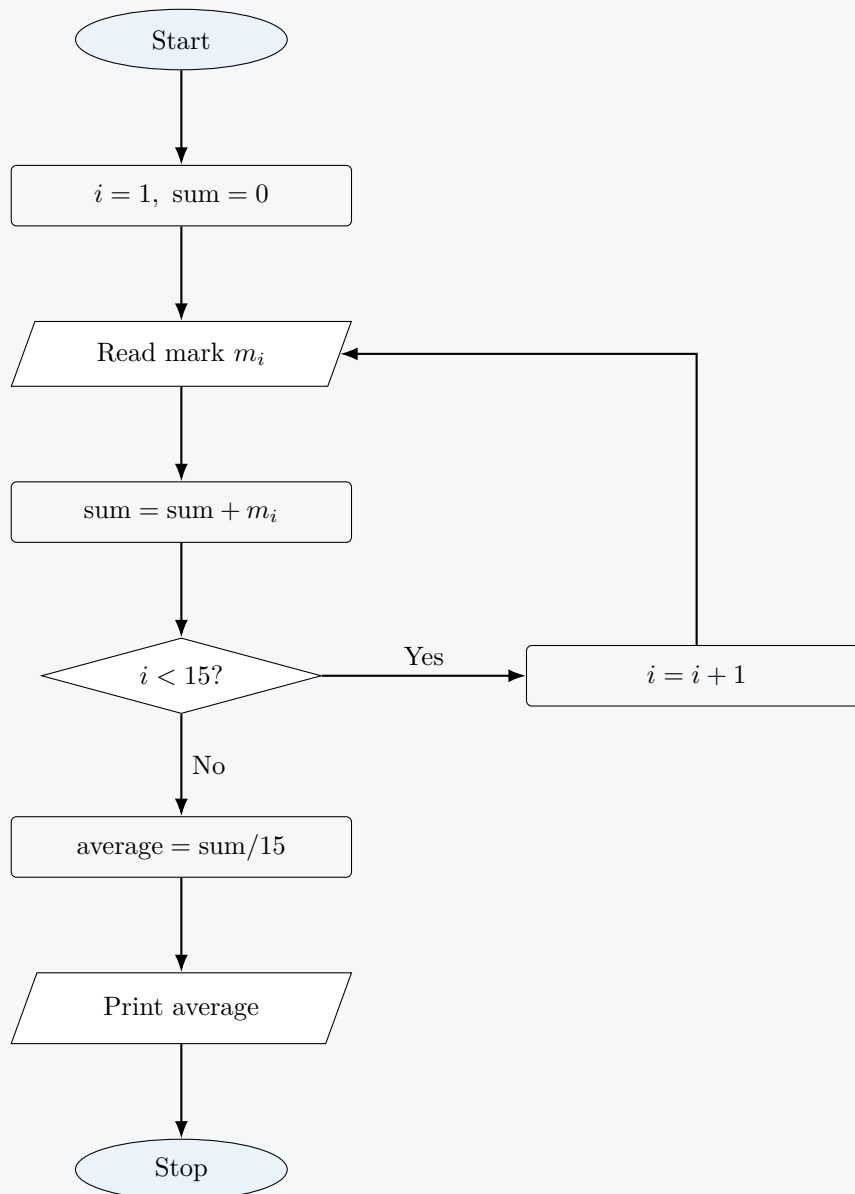
Answer. The required software is a **printer driver**. A driver is a system software component that allows the operating system and applications to communicate correctly with a hardware device.

Features of a printer driver:

- It translates print commands into a format understood by the printer.
- It provides settings such as paper size, print quality, colour/black-and-white mode and duplex printing.
- It helps the operating system detect, install and manage the printer.
- It may include utilities for printer status, ink/toner level, alignment and troubleshooting.

Question. (e) Draw a flowchart for a program to find the average marks of the students of a class in a subject. The class strength is 15. [5 marks]

Answer. The program reads marks of 15 students, adds them, and divides the total by 15.



Question. (f) List six operations that can be performed using a word processor.

[3 marks]

Answer. Six common operations of a word processor are:

1. Creating and editing text documents.
2. Formatting text using font, size, bold, italic, underline and colour.
3. Page formatting such as margins, headers, footers and page numbers.
4. Spell-checking and grammar-checking.
5. Inserting tables, pictures, symbols and charts.
6. Printing, saving, exporting and sharing documents.

Question. (g) Explain the characteristics of a hub and a router. Which of the two will be needed if you want to connect from one network to another?

[5 marks]

Answer. A **hub** is a simple networking device that connects multiple computers within the same local network. It broadcasts incoming data to all connected ports, so every device receives the data whether it is the intended receiver or not. It works mainly at the physical layer, has no intelligence for path selection, and can cause unnecessary traffic.

A **router** connects two or more different networks. It reads logical addresses such as IP addresses and forwards packets to the correct destination network. It can select routes, separate broadcast domains, connect a LAN to the Internet, and may also provide NAT, firewall and DHCP services.

To connect from **one network to another**, a **router** is needed, because a hub cannot route packets between different networks.

Question. (h) What is an IPv4 address? What are its components? What is the use of a subnet mask? Explain with the help of an example. [6 marks]

Answer. An **IPv4 address** is a 32-bit logical address used to identify a device on an IP network. It is usually written in dotted decimal form as four octets, for example:

192.168.1.25

Each octet is 8 bits and can range from 0 to 255.

Components of IPv4 address:

- **Network part:** identifies the network to which the device belongs.
- **Host part:** identifies the particular device within that network.

A **subnet mask** tells which bits of the IP address represent the network part and which bits represent the host part. It is also 32 bits. A binary 1 in the mask represents the network portion; a binary 0 represents the host portion.

Example:

IP address = 192.168.1.25, subnet mask = 255.255.255.0.

Here, the first three octets form the network part:

Network = 192.168.1.0, Host = 25.

So all devices from 192.168.1.1 to 192.168.1.254 generally belong to the same subnet, while 192.168.2.10 belongs to a different subnet.

Question. (i) What is a browser? Why is it needed? List any two features of a browser. [4 marks]

Answer. A **web browser** is an application software used to access, retrieve and display web pages from the Internet or a local network. Examples include Chrome, Firefox, Edge and Safari.

It is needed because web pages are written using languages and protocols such as HTML, CSS, JavaScript, HTTP and HTTPS. The browser interprets these and presents the information in a readable and interactive form.

Two features of a browser:

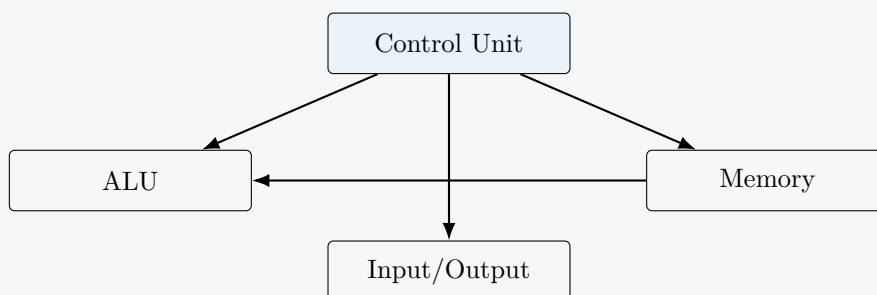
- Tabbed browsing, bookmarks and history for easy navigation.
- Security features such as HTTPS support, pop-up control, safe browsing warnings and password management.

2 Question 2

Question. (a) What is the role of Arithmetic Logic Unit (ALU) and Control Unit (CU) in a computer? [4 marks]

Answer. The **ALU** and **CU** are two major parts of the CPU.

- **Arithmetic Logic Unit (ALU):** Performs arithmetic operations such as addition, subtraction, multiplication and division. It also performs logical operations such as comparison, AND, OR and NOT.
- **Control Unit (CU):** Controls and coordinates all operations of the computer. It fetches instructions from memory, decodes them, directs the ALU, manages data movement and controls input/output operations.



Question. (b) What is a supercomputer? How is it different from a mainframe computer? [4 marks]

Answer. A **supercomputer** is an extremely powerful computer designed to perform very high-speed calculations and process very large scientific or engineering problems. It is used in weather forecasting, climate modelling, space research, nuclear simulations, bioinformatics and artificial intelligence.

Basis	Supercomputer	Mainframe computer
Main purpose	Maximum computational speed	Reliable processing of huge transactions
Typical use	Scientific simulations, research, modelling	Banking, railways, airlines, government records
Processing style	Parallel numerical computation	Multi-user transaction processing
Performance measure	FLOPS and parallel processing capability	I/O capacity, reliability and number of users supported

Question. (c) What are the uses of magnetic tape? Explain how the access mechanism of magnetic tape is different from that of magnetic disks. [4 marks]

Answer. Uses of magnetic tape:

- Backup of large amounts of data.
- Archival storage where data is kept for long periods.
- Storing rarely used data at low cost.
- Offline storage and disaster recovery.

Access mechanism difference: Magnetic tape is a **sequential access** medium. To reach a particular record, the tape may need to move through earlier records first. Magnetic disk is a **direct/random access** medium. The read/write head can move to the required track and sector directly, so retrieval is much faster.

Thus, magnetic tape is cheaper for backup and archiving, while magnetic disk is better for frequently accessed data.

Question. (d) Explain the terms pixel, resolution and LEDs in the context of computer monitors.
[4 marks]

Answer.

- **Pixel:** The smallest addressable picture element on a monitor screen. Images and text are displayed by controlling many pixels.
- **Resolution:** The number of pixels displayed horizontally and vertically. For example, 1920×1080 means 1920 pixels across and 1080 pixels down. Higher resolution generally gives sharper display.
- **LEDs:** Light Emitting Diodes. In LED monitors, LEDs are used as a backlight source for the display. They consume less power, provide better brightness and are thinner than older monitor technologies.

Question. (e) Explain the role of Recycle Bin and Taskbar on Windows operating system.
[4 marks]

Answer.

- **Recycle Bin:** It is a temporary storage area for deleted files and folders. If a file is deleted accidentally, it can be restored from the Recycle Bin. When the Recycle Bin is emptied, the files are permanently removed from normal user access.
- **Taskbar:** It is the horizontal bar usually present at the bottom of the Windows desktop. It shows the Start button, pinned apps, currently running applications, notification area, date/time and system icons. It helps users switch quickly between open programs.

3 Question 3

Question. (a) What are various components of a Local Area Network (LAN)? What are the advantages of using LANs? How are LANs different from WANs?
[6 marks]

Answer. Components of LAN:

- Computers/workstations and servers.
- Network Interface Cards (NICs).
- Switches/hubs and routers.
- Transmission media such as twisted pair cable, fibre optic cable or Wi-Fi.
- Network operating system, protocols and shared resources such as printers and storage.

Advantages of LAN:

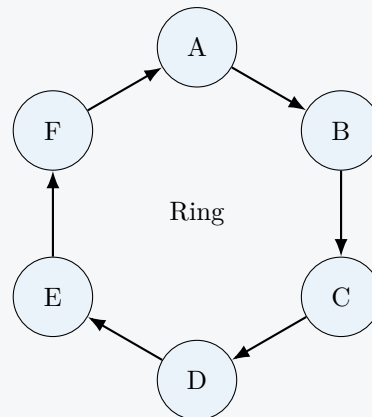
- Sharing of files, printers, Internet connection and software resources.
- Fast communication within a small area.
- Centralized data management and backup.
- Lower cost of resource sharing.
- Easier administration and security control.

LAN versus WAN:

Basis	LAN	WAN
Coverage	Small area such as room, office, campus	Large area such as city, country or world
Ownership	Usually owned by one organization	May use telecom/ISP infrastructure
Speed	Usually high	Usually lower than LAN over long distance
Cost	Lower setup and operation cost	Higher cost
Example	College computer lab network	Internet, bank branch network

Question. (b) Explain the ring topology with the help of a diagram. List the advantages and disadvantages of ring topology. *[6 marks]*

Answer. In **ring topology**, each computer is connected to exactly two neighbouring computers, forming a closed circular path. Data travels from one node to the next around the ring until it reaches the destination. In many ring systems, a token is used to control transmission and avoid collisions.



Advantages:

- Data collision is reduced because transmission can be controlled by token passing.
- Performance is predictable under heavy traffic.
- Each node gets a fair chance to transmit.

Disadvantages:

- Failure of one node or cable can affect the whole network unless a dual ring or bypass mechanism is used.
- Adding or removing nodes is difficult compared with star topology.
- Troubleshooting can be time-consuming.

Question. (c) Explain the following web applications, giving their features and advantages: (i) E-learning, (ii) E-mail. [8 marks]

Answer. (i) E-learning: E-learning means learning through electronic technologies, especially the Internet. It may include online classes, recorded lectures, learning management systems, quizzes, discussion forums and digital notes.

Features:

- Online delivery of study material, videos and assignments.
- Self-paced learning and repeated viewing of content.
- Online tests, discussion forums and feedback.
- Access through computer, tablet or mobile phone.

Advantages:

- Flexible learning from any place and at any time.
- Reduces travel and printed-material cost.
- Useful for distance education and working learners.
- Supports multimedia learning through text, audio, video and animation.

(ii) E-mail: E-mail means electronic mail. It is a web/network application used to send and receive digital messages through the Internet.

Features:

- Send text messages, attachments, images and documents.
- Supports CC, BCC, reply, forward and mailing lists.
- Provides inbox, sent, drafts, spam filter and search facility.
- Can be accessed through browser or mail client.

Advantages:

- Fast and low-cost communication.
- Same message can be sent to many people.
- Provides written record of communication.
- Attachments make document sharing easy.

4 Question 4

Question. (a) What is a computer virus? What role can anti-virus software play to control a computer virus? *[4 marks]*

Answer. A **computer virus** is a malicious program that attaches itself to files, programs or boot sectors and spreads when the infected item is executed or shared. It may corrupt data, slow down the system, steal information, display unwanted messages or damage software.

Role of antivirus software:

- Scans files, memory and storage devices for known malware signatures.
- Detects suspicious behaviour using heuristic methods.
- Quarantines, repairs or deletes infected files.
- Provides real-time protection while downloading, copying or opening files.
- Updates virus definitions to detect new threats.

Question. (b) What are open source software? What are the primary objectives of the open source movement? [4 marks]

Answer. **Open source software** is software whose source code is made available to users so that they can study, use, modify and distribute it under the terms of an open source licence. Examples include Linux, LibreOffice, Firefox and Apache.

Primary objectives of open source movement:

- Promote freedom to study and modify software.
- Encourage collaborative development and peer review.
- Reduce dependence on proprietary software vendors.
- Improve transparency, security and innovation.
- Make software affordable and accessible to more users.

Question. (c) What is Graphical User Interface? What are the advantages of using these interfaces? [4 marks]

Answer. A **Graphical User Interface (GUI)** is a user interface that allows users to interact with the computer using graphical elements such as windows, icons, menus, buttons, dialog boxes and pointer devices instead of only typed commands.

Advantages:

- Easy to learn and use, especially for beginners.
- Reduces the need to memorize commands.
- Supports visual interaction through icons and menus.
- Allows multitasking through multiple windows.
- Makes file management, printing and application use more convenient.

Question. (d) Explain the terms subroutine and function in the context of programming languages with the help of an example each. [5 marks]

Answer. A **subroutine** is a named block of code written to perform a specific task. It can be called whenever that task is required. A subroutine may or may not return a value, depending on the language.

Example of subroutine/procedure:

```
Procedure PrintLine()
    Print "-----"
End Procedure
```

This subroutine prints a line when called.

A **function** is also a named block of code, but it usually returns a value to the calling program.

Example of function:

```
Function Square(x)
    Return x * x
End Function
```

If **Square(5)** is called, it returns 25.

Thus, a function is mainly used when a computed value has to be returned, while a subroutine is mainly used to perform an action.

Question. (e) What is meant by macros in the context of spreadsheet software? Why are they needed? *[3 marks]*

Answer. In spreadsheet software, a **macro** is a recorded or written sequence of commands used to automate repeated tasks. Macros are often written in a scripting language such as VBA in Microsoft Excel or recorded through the macro recorder.

Need for macros:

- To automate repetitive formatting, calculation or reporting tasks.
- To save time and reduce manual errors.
- To perform complex operations with a single command or button.
- To standardize repeated work such as monthly reports.

5 Question 5

Question. Discuss the uses of any eight of the following in a computer system/networking. The solution below gives all twelve for complete preparation. *[20 marks]*

Answer.

- (i) **Firewall:** A firewall is a security system that monitors and controls incoming and outgoing network traffic. It allows trusted traffic and blocks unauthorized or harmful traffic. It protects a network from attacks, malware communication and unauthorized access.
- (ii) **TCP/IP Protocol:** TCP/IP is the basic protocol suite of the Internet. IP handles addressing and routing of packets, while TCP provides reliable delivery by sequencing, acknowledgement and retransmission. It enables computers on different networks to communicate.
- (iii) **Radio waves:** Radio waves are used in wireless communication such as Wi-Fi, Bluetooth, mobile networks and wireless keyboards/mice. They allow data transmission without physical cables.
- (iv) **Twisted Pair Cable:** Twisted pair cable is a common guided transmission medium used in LANs and telephone networks. The twisting reduces electromagnetic interference. UTP cables such as Cat5e and Cat6 are widely used for Ethernet connections.
- (v) **Compiler:** A compiler translates a complete high-level language program into machine code or object code before execution. It detects syntax errors and produces executable code. Languages such as C and C++ commonly use compilers.
- (vi) **Linker:** A linker combines object files, library files and required modules to produce a final executable program. It resolves references between modules, such as function calls defined in another file or library.
- (vii) **Presentation Software:** Presentation software is used to create slide-based visual presentations. It supports text, images, charts, tables, animations and multimedia. Examples include Microsoft PowerPoint, LibreOffice Impress and Google Slides.
- (viii) **Database Management:** Database management systems store, organize, retrieve and manage large amounts of structured data. They support queries, updates, reports, data security, backup and multi-user access. Examples include MySQL, Oracle and MS Access.
- (ix) **ASCII:** ASCII stands for American Standard Code for Information Interchange. It is a character encoding scheme that represents letters, digits, punctuation marks and control characters using numeric codes. It helps computers store and exchange text.
- (x) **DVD:** DVD stands for Digital Versatile Disc or Digital Video Disc. It is an optical storage medium used to store software, videos, backups and data files. It has more storage capacity than a CD.
- (xi) **Mouse:** A mouse is a pointing input device used to move the pointer on the screen, select objects, open files, drag items and interact with graphical user interfaces. It usually has buttons and a scroll wheel.
- (xii) **Bar code reader:** A bar code reader is an input device that scans printed bar codes and converts them into digital data. It is used in shops, libraries, warehouses and billing systems for quick identification of products or items.

Quick revision summary

- Number conversion requires grouping binary bits in powers of 2 or groups of four for hexadecimal.
- RAM is volatile working memory; ROM is non-volatile firmware memory.
- A printer needs a device driver.
- A router connects different networks; a hub only connects devices in the same local segment.
- IPv4 has network and host parts, separated using a subnet mask.

- Ring topology forms a closed loop; it has predictable access but difficult fault handling.
- Antivirus, firewall, open source software, compiler, linker and database management are core practical terms.